

The pizzetti proofs, explained

Part 1: The interior series and its error bound

Companion to the pizzetti paper (Sections 2–4 and Appendix A)

Abstract

This is a slow, step-by-step version of the proof that the pizzetti interior series has a guaranteed error bound (Theorem 2 of the paper). It assumes only first-year calculus: the binomial series, geometric series, and averaging a function over a disk. Every step comes with a small numerical example, and every formula is checked independently in the validation notebooks of the repository (`validation/02.series.theorem2.ipynb`). The paper itself is the authoritative, terse version; nothing here goes beyond it.

Where is what. The three parts of this companion follow the paper:

Paper	Content	Here
Sections 2–3, 4.1, Appendix A	interior series, Theorem 2, the cut	Part 1
Section 4.2, Appendix B (incl. B.1)	mean-value expansion, Propositions 1–2	Part 2
Sections 5.1, 5.3, Appendix C	contact expansions, Theorem 3	Part 3

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1 The question

A star is a disk of radius 1. Its brightness drops towards the edge: at distance r from the centre the surface brightness is a sum of powers of

$$\mu = \sqrt{1 - r^2},$$

for example $I = 1 - u_1(1 - \mu) - u_2(1 - \mu)^2$ for the quadratic law. A planet is a dark disk of radius p (in units of the stellar radius) whose centre lies at distance z from the stellar centre

(Figure 1). While the planet lies *entirely inside* the star, that is $z + p < 1$, the light it blocks is the brightness averaged over the planet's disk, times its area. Every power μ^α is $(1 - r^2)^\gamma$ with $\gamma = \alpha/2$, so everything reduces to one quantity, the *disk average*

$$D_\gamma(z, p) = \langle (1 - r^2)^\gamma \rangle = \frac{1}{\pi p^2} \int_{\text{planet disk}} (1 - r^2)^\gamma dA. \quad (1)$$

For whole numbers γ this is a polynomial and easy. For $\gamma = \frac{1}{2}$ (the μ term of the quadratic law) it needs elliptic integrals, and for other values such as $\gamma = 0.3$ there is no closed formula at all. The paper replaces D_γ by a short polynomial and proves how large the error can be. This part explains that proof.

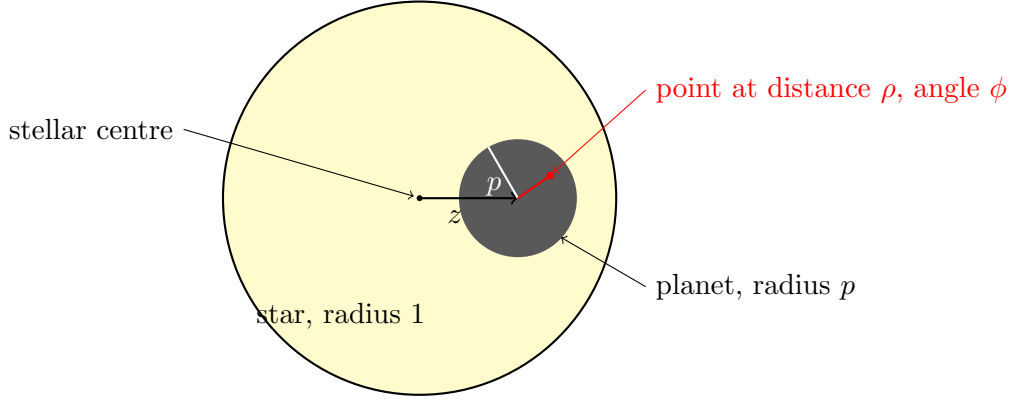


Figure 1: A planet of radius p at distance z from the stellar centre. A point of the planet's disk is described by its distance $\rho \leq p$ from the planet's centre and an angle ϕ .

2 Step 1: Pull out the value at the planet's centre

A point of the planet's disk at distance ρ from the planet's centre, at angle ϕ , has squared distance from the stellar centre

$$r^2 = z^2 + 2z\rho \cos \phi + \rho^2$$

(the law of cosines). Write $w = 1 - z^2$: this is $1 - r^2$ at the planet's centre. Then

$$1 - r^2 = w - (2z\rho \cos \phi + \rho^2) = w(1 - \varepsilon), \quad \varepsilon = \frac{2z\rho \cos \phi + \rho^2}{w}. \quad (2)$$

So $(1 - r^2)^\gamma = w^\gamma(1 - \varepsilon)^\gamma$. The factor w^γ is the brightness at the planet's centre; ε measures how much the brightness at the point differs from it.

How large can ε be? Since $|\cos \phi| \leq 1$ and $\rho \leq p$,

$$|\varepsilon| \leq \varepsilon_{\max} = \frac{2zp + p^2}{1 - z^2}. \quad (3)$$

And $\varepsilon_{\max} < 1$ exactly when $2zp + p^2 < 1 - z^2$, i.e. $(z + p)^2 < 1$, i.e. when the planet lies inside the star. That is the condition we need below.

Worked example. Throughout this part we use $\gamma = \frac{1}{2}$, $p = 0.1$, $z = 0.5$. Then $w = 0.75$ and $\varepsilon_{\max} = (0.1 + 0.01)/0.75 = 0.14\bar{6}$.

3 Step 2: Expand in powers of ε

For $|\varepsilon| < 1$ the binomial series gives

$$(1 - \varepsilon)^\gamma = \sum_{m=0}^{\infty} \beta_m \varepsilon^m, \quad \beta_m = (-1)^m \binom{\gamma}{m} = \frac{(-1)^m}{m!} \gamma(\gamma-1) \cdots (\gamma-m+1). \quad (4)$$

For $\gamma = \frac{1}{2}$ this is the familiar $\sqrt{1-\varepsilon} = 1 - \frac{1}{2}\varepsilon - \frac{1}{8}\varepsilon^2 - \frac{1}{16}\varepsilon^3 - \frac{5}{128}\varepsilon^4 - \dots$
Two facts about the coefficients carry the whole proof.

Fact 1 (one sign). Going from β_m to β_{m+1} multiplies by

$$\frac{\beta_{m+1}}{\beta_m} = -\frac{\gamma-m}{m+1} = \frac{m-\gamma}{m+1}. \quad (5)$$

Once $m > \gamma$ this ratio is positive, so from there on *all coefficients have the same sign*. For $0 < \gamma < 1$ that happens already at $m = 1$: every β_m with $m \geq 1$ is negative (as in the $\sqrt{1-\varepsilon}$ series above). For a general non-integer γ the common sign is $(-1)^{\lceil \gamma \rceil}$, where $\lceil \gamma \rceil$ is γ rounded up.

Fact 2 (shrinking). For $m > \gamma$ the ratio (5) is also less than 1, because $m - \gamma < m + 1$. So $|\beta_{m+1}| < |\beta_m|$: the coefficients shrink.

(For whole numbers γ the factor $\gamma - m$ eventually vanishes and the series stops; that is why integer exponents are easy.)

4 Step 3: Average each power over the disk

We now average term by term over the planet's disk:

$$D_\gamma = w^\gamma \sum_{m=0}^{\infty} \beta_m \langle \varepsilon^m \rangle. \quad (6)$$

Why we may average term by term. On the disk $|\varepsilon| \leq \varepsilon_{\max} < 1$. Let m_0 be the first index above γ ; from there on the coefficients shrink (Fact 2), so every term with $m \geq m_0$ is at most $|\beta_{m_0}| \varepsilon_{\max}^m$ in size. The tail of the series after m terms is therefore smaller than the geometric tail $|\beta_{m_0}| \varepsilon_{\max}^m / (1 - \varepsilon_{\max})$ *at every point of the disk at once*. Averaging a sum whose tail is uniformly small is the same as summing the averages. (In the language of analysis: the series converges uniformly, so it can be integrated term by term.)

The averages $\langle \varepsilon^m \rangle$ are simple. Expand $(2z\rho \cos \phi + \rho^2)^m$ with the binomial theorem and average over the disk, where $dA = \rho d\rho d\phi$. Two elementary averages are needed:

- the average of $\cos^l \phi$ over a full turn is $\binom{l}{l/2} 2^{-l}$ for even l (for instance $\frac{1}{2}$ for $l = 2$, $\frac{3}{8}$ for $l = 4$) and 0 for odd l ;
- the average of ρ^n over the disk is $2\rho^n / (n + 2)$.

Collecting terms gives Lemma 1 of the paper:

$$\langle \varepsilon^m \rangle = \frac{1}{w^m} \sum_{\substack{l=0 \\ l \text{ even}}}^m G_{m,l} z^l p^{2m-l}, \quad G_{m,l} = \binom{m}{l} \binom{l}{l/2} \frac{2}{2m-l+2}. \quad (7)$$

For example

$$\langle \varepsilon \rangle = \frac{p^2/2}{w}, \quad \langle \varepsilon^2 \rangle = \frac{z^2 p^2 + p^4/3}{w^2}.$$

(Check the second one by hand: $\varepsilon^2 w^2 = 4z^2 \rho^2 \cos^2 \phi + 4z \rho^3 \cos \phi + \rho^4$; the middle term averages to zero, $\langle \rho^2 \rangle = p^2/2$, $\langle \cos^2 \phi \rangle = \frac{1}{2}$ and $\langle \rho^4 \rangle = p^4/3$.)

Fact 3 (never negative). Only even powers of $\cos \phi$ survive the averaging, and every surviving term in (7) is positive. So

$$\langle \varepsilon^m \rangle \geq 0 \quad \text{for every } m.$$

This is not obvious from the start, since ε itself is negative on part of the disk (where $\cos \phi < 0$).

Worked example. For $\gamma = \frac{1}{2}$, $p = 0.1$, $z = 0.5$ the terms $w^\gamma \beta_m \langle \varepsilon^m \rangle$ of (6) are

m	0	1	2	3	4	5
term	0.866025	-2.89×10^{-3}	-4.88×10^{-4}	-6.45×10^{-6}	-1.42×10^{-6}	-4.78×10^{-8}

They fall off quickly, and after the first all have the same sign (Facts 1 and 3). The exact value, from the defining integral, is $D_{1/2} = 0.86264319100026 \dots$

5 Step 4: Cut the series off and bound what is left

Keep the terms up to $m = M$:

$$S_M = w^\gamma \sum_{m=0}^M \beta_m \langle \varepsilon^m \rangle.$$

What is left over is the *remainder*

$$D_\gamma - S_M = w^\gamma \sum_{m=M+1}^{\infty} \beta_m \langle \varepsilon^m \rangle.$$

Theorem 2 (as in the paper). Let $z + p < 1$, let $\gamma > 0$ not be a whole number, and let M be odd with $M + 1 > \gamma$. Then

$$|D_\gamma - S_M| \leq B_M = \frac{w^\gamma |\beta_{M+1}| \langle \varepsilon^{M+1} \rangle}{1 - \varepsilon_{\max}}, \quad (8)$$

and the remainder $D_\gamma - S_M$ has the sign $(-1)^{\lceil \gamma \rceil}$.

Proof, one idea at a time. (a) All remainder terms have the same sign. Every m in the remainder satisfies $m \geq M + 1 > \gamma$, so all β_m there have the common sign $(-1)^{\lceil \gamma \rceil}$ (Fact 1), and all $\langle \varepsilon^m \rangle \geq 0$ (Fact 3). A sum of terms of one sign has that sign, and its size is the sum of the sizes:

$$|D_\gamma - S_M| = w^\gamma \sum_{m \geq M+1} |\beta_m| \langle \varepsilon^m \rangle.$$

(b) Replace each coefficient by the first one. The coefficients shrink (Fact 2), so $|\beta_m| \leq |\beta_{M+1}|$ for all $m \geq M + 1$.

(c) Compare each average with the first one. Here is where M odd is used. Since $M + 1$ is even, $\varepsilon^{M+1} = |\varepsilon|^{M+1}$. For $m \geq M + 1$, at every point of the disk,

$$\varepsilon^m \leq |\varepsilon|^m = |\varepsilon|^{M+1} |\varepsilon|^{m-M-1} \leq \varepsilon^{M+1} \varepsilon_{\max}^{m-M-1},$$

and averaging keeps the inequality: $\langle \varepsilon^m \rangle \leq \varepsilon_{\max}^{m-M-1} \langle \varepsilon^{M+1} \rangle$.

(d) Add up. Putting (b) and (c) into (a),

$$|D_\gamma - S_M| \leq w^\gamma |\beta_{M+1}| \langle \varepsilon^{M+1} \rangle (1 + \varepsilon_{\max} + \varepsilon_{\max}^2 + \dots) = \frac{w^\gamma |\beta_{M+1}| \langle \varepsilon^{M+1} \rangle}{1 - \varepsilon_{\max}},$$

the sum of a geometric series with ratio $\varepsilon_{\max} < 1$. □

Why the bound is so close to the true error. The bound is the first omitted term divided by $1 - \varepsilon_{\max}$. The true remainder is the first omitted term *plus* smaller terms of the same sign. When the terms fall off quickly, both are close to the first omitted term.

Worked example. For $\gamma = \frac{1}{2}$, $p = 0.1$, $z = 0.5$:

M	$D_{1/2} - S_M$	B_M	error/bound
1	-4.95×10^{-4}	5.71×10^{-4}	0.87
3	-1.47×10^{-6}	1.66×10^{-6}	0.89
5	-9.50×10^{-9}	1.05×10^{-8}	0.91

The remainder is negative, as predicted by the sign $(-1)^{\lceil 1/2 \rceil} = -1$, and always below the bound.

Check it yourself. Notebook `validation/02_series_theorem2.ipynb`, cell 2b, repeats this for several exponents, radius ratios and separations against the defining integral (1), computed by brute-force numerical integration. In the full run (2132 cases) no remainder exceeded its bound, every sign was as predicted, and the largest ratio of error to bound was 0.99998.

6 Step 5: One cut per light curve

The bound grows as the planet moves outwards. Write the bound using (7) as

$$B_M = |\beta_{M+1}| w^{\gamma-M-1} \sum_l G_{M+1,l} z^l p^{2M+2-l} \frac{1}{1 - \varepsilon_{\max}}.$$

As z increases, $w = 1 - z^2$ decreases, so $w^{\gamma-M-1}$ (a negative power) increases; each z^l increases; and $\varepsilon_{\max}$ increases, so $1/(1 - \varepsilon_{\max})$ increases. Every factor grows, hence so does B_M .

Consequence. For a given planet (p), order M and tolerance τ , there is a largest separation z_c up to which the bound stays below τ , and below z_c it is below τ *everywhere*. One number per light curve decides where the series may be used (Corollary 1 of the paper). For the quadratic law the flux error is the bound times a constant factor $p^2 \Omega |u_1 + 2u_2|$ that depends only on the limb-darkening coefficients.

Worked example. Quadratic law $(u_1, u_2) = (0.4, 0.25)$, $M = 11$, flux tolerance 10^{-9} : for an Earth-sized planet ($p = 0.0092$) the cut lies at $z_c = 0.99767(1 - p)$, for $p = 0.1$ at $z_c = 0.912(1 - p)$. Beyond z_c , near the stellar edge, other methods take over (Parts 2 and 3).

Check it yourself. Notebook `validation/02_series_theorem2.ipynb`, cell 2c, evaluates pizzetti's series on dense grids $0 \leq z \leq z_c$ and compares with exoplanet-core, an implementation of the exact solution of Agol, Luger & Foreman-Mackey (2020). The error never exceeded the tolerance.

7 Step 6: Why it is fast — the polynomial form

Each $\langle \varepsilon^m \rangle$ is a polynomial in z^2 and p^2 divided by w^m . Substituting $z^2 = 1 - 1/v$ with $v = 1/w$ turns S_M into

$$S_M = w^\gamma \sum_{j=0}^M a_j(p) v^j,$$

a polynomial of degree M in v whose coefficients a_j depend only on the planet size p (Equations (13)–(14) of the paper give them explicitly). For $\gamma = \frac{1}{2}$, $M = 3$:

$$S_3 = \sqrt{w} \left[1 - \frac{p^2}{8}(v + v^2) + \frac{p^4}{12}v^2 - \frac{p^4}{8}v^3 - \frac{p^6}{64}v^3 \right].$$

For a light curve the a_j are computed once. Each data point then costs one subtraction for w , one division for v , one square root (or power) for w^γ , and M multiply-adds. That is the source of the speed-up over elliptic integrals.

Check it yourself. Notebook `validation/02_series_theorem2.ipynb`, cell 2a: the polynomial form equals the partial sum to 40 digits, and the S_3 formula above is exact.

8 Summary

The whole argument uses four elementary facts: the brightness can be written as its value at the planet’s centre times $(1 - \varepsilon)^\gamma$; the binomial coefficients eventually have one sign and shrink; the disk averages of ε^m are never negative; and a geometric series sums to $1/(1 - \text{ratio})$. Together they give an error bound that is never exceeded, is close to the true error, and grows with z , so a single cut per light curve guarantees the requested accuracy.

What the proof does *not* cover is rounding in double-precision arithmetic, which is checked separately (it stays below 10^{-16} of the stellar flux) and is negligible compared with the tolerances used.

Next: Part 2 relates the series to Pizzetti’s mean-value expansion and proves two further bounds (Appendix B of the paper). Part 3 treats the phases in which the planet crosses the stellar edge (Section 5.3 and Appendix C).